

Linear Algebra for Quantum Computation

Lecture Notes

The purpose of these notes is to compile the definitions, notations, and properties of linear algebra that are important for quantum computation. They also serve as a quick reference for the main operations in vector spaces, for instance, the inner and tensor products. Quantum computation inherited linear algebra from quantum mechanics as the supporting language for describing this area. Therefore, it is essential to have a solid knowledge of the basic results of linear algebra to understand quantum computation and quantum algorithms.

The material is organised as two lectures with two blocks each. Every block opens with a list of learning objectives and closes with the relevant exercises. All exercises are in [Appendix A](#) and their solutions in [Appendix B](#).

Lecture	Block	Contents
1	1	Vector spaces, inner product, Dirac notation, computational basis
1	2	The qubit and the Bloch sphere, linear operators, matrix representation
2	3	Diagonal representation, completeness, Cauchy–Schwarz, special operators, Pauli matrices
2	4	Operator functions, tensor product, registers

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Notation Cheat Sheet

Symbol	Reads as	Meaning
$\mathbb{C}^n$	—	complex vector space of column vectors with n entries
$\mathcal{H}^N$	—	N -dimensional Hilbert space; $\mathcal{H}_A$: space of system A
$ v\rangle$	“ket v ”	a vector, written as a column matrix
$\langle v $	“bra v ”	the dual vector, a row matrix, $\langle v = [a_1^* \cdots a_n^*]$
$\langle v w\rangle$	“bra-ket”	inner product, a complex number, $\sum_i a_i^* b_i$
$ v\rangle\langle w $	—	outer product, an $n \times n$ matrix (an operator)
$ v\rangle\langle v $	—	projector onto $ v\rangle$ when $\ v\ = 1$
$\langle v A w\rangle$	—	matrix element of A ; equals $\langle v Aw\rangle$
$\ v\ $	“norm”	$\sqrt{\langle v v\rangle}$
δ_{ij}	“Kronecker delta”	1 if $i = j$, 0 otherwise
$ 0\rangle, \dots, n-1\rangle$	—	computational (canonical) basis of $\mathbb{C}^n$
$ D\rangle$	“diagonal state”	$\frac{1}{\sqrt{n}} \sum_{i=0}^{n-1} i\rangle$; for $n = 2$, $ D\rangle = +\rangle$
$ \pm\rangle$	—	$(0\rangle \pm 1\rangle)/\sqrt{2}$
A^*	“conjugate”	entrywise complex conjugate
A^T	“transpose”	$(A^T)_{ij} = A_{ji}$
$A^\dagger$	“ A dagger”	adjoint = $(A^*)^T$
$\text{tr}(A)$	“trace”	$\sum_i a_{ii} = \sum_i \langle v_i A v_i\rangle$
$\otimes$	“tensor”	tensor product; Kronecker product on matrices
$V^{\otimes n}$	—	$V \otimes \cdots \otimes V$ with n terms
$\oplus$	“xor”	bitwise binary sum
$ i, j\rangle$	—	shorthand for $ i\rangle \otimes j\rangle$

Operator class	Definition	An important consequence
normal	$A^\dagger A = AA^\dagger$	diagonalisable in an orthonormal basis
unitary	$U^\dagger U = UU^\dagger = I$	normal; eigenvalues $e^{i\alpha}$; preserves inner products
Hermitian	$A^\dagger = A$	normal; real eigenvalues; orthogonal eigenspaces
projector	$P^2 = P, P^\dagger = P$	Hermitian; eigenvalues 0, 1; $I - P$ also a projector
positive	$\langle v A v\rangle \geq 0 \ \forall v\rangle$	Hermitian; eigenvalues ≥ 0
positive def.	$\langle v A v\rangle > 0 \ \forall v\rangle \neq \mathbf{0}$	eigenvalues > 0 ; invertible

$$\sigma_0 = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \sigma_1 = \sigma_x = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \sigma_2 = \sigma_y = Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad \sigma_3 = \sigma_z = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\sigma_j^2 = I, \quad \sigma_j^\dagger = \sigma_j, \quad \text{eigenvalues } \pm 1, \quad X|0\rangle = |1\rangle, X|1\rangle = |0\rangle, \quad Z|0\rangle = |0\rangle, Z|1\rangle = -|1\rangle$$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle, \quad H^2 = I$$

Identity	Statement
completeness	$I = \sum_i v_i\rangle\langle v_i $ for any orthonormal basis $\{ v_i\rangle\}$
matrix from an operator	$A = \sum_{i,j} a_{ij} w_i\rangle\langle v_j , \quad a_{ij} = \langle w_i A v_j\rangle$
diagonal representation	$O = \sum_i \lambda_i v_i\rangle\langle v_i = \sum_\lambda \lambda P_\lambda, \quad \sum_\lambda P_\lambda = I$
operator function	$f(A) = \sum_i f(a_i) v_i\rangle\langle v_i $ for normal A
Cauchy–Schwarz	$ \langle v w\rangle \leq \sqrt{\langle v v\rangle} \sqrt{\langle w w\rangle}$
trace, cyclic	$\text{tr}(AB) = \text{tr}(BA), \quad \text{tr}(ABC) = \text{tr}(CAB)$
trace with a projector	$\text{tr}(A \psi\rangle\langle \psi) = \langle \psi A \psi\rangle$
mixed product	$(A_1 \otimes B_1)(A_2 \otimes B_2) = (A_1 A_2) \otimes (B_1 B_2)$
tensor dagger	$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$
tensor trace / det	$\text{tr}(A \otimes B) = \text{tr } A \text{ tr } B, \quad \det(A \otimes B) = (\det A)^m (\det B)^n$
tensor norm	$\ v\rangle \otimes w\rangle \ = \ v\rangle \ \cdot \ w\rangle \ $
Bloch parametrisation	$ \psi\rangle = \cos \frac{\theta}{2} 0\rangle + e^{i\phi} \sin \frac{\theta}{2} 1\rangle$
Hadamard, n qubits	$\langle i H^{\otimes n} j\rangle = \frac{(-1)^{i \cdot j}}{\sqrt{2^n}}$

Part I

Lecture 1 Vectors, Dirac Notation, Qubits and Operators

Quantum computation inherited linear algebra from quantum mechanics as the supporting language for describing this area. Therefore, it is essential to have a solid knowledge of the basic results of linear algebra to understand quantum computation and quantum algorithms. This first lecture builds the vocabulary: vector spaces over $\mathbb{C}$, the inner product, Dirac's bra-ket notation, the computational basis, the qubit, and linear operators together with their matrix representations.

1 Block 1: Vector Spaces, Inner Product, Dirac Notation

Learning goals:

- state the axioms of a complex vector space and check them on $\mathbb{C}^n$;
- compute inner products, norms and orthogonality relations in $\mathbb{C}^n$, and explain why the inner product is *conjugate*-linear in its first argument;
- translate freely between ordinary vector notation, matrix notation and Dirac notation, and read a string of bras and kets as a matrix product;
- compute inner and outer products in Dirac notation and recognise which one produces a number and which one produces an operator;
- write down the computational basis of $\mathbb{C}^n$ and the diagonal state $|D\rangle$.

A short raison d'être

The state of an isolated quantum system is a unit vector in a complex Hilbert space; the operations a quantum computer performs are unitary operators on that space; a measurement is described by projectors; and putting two systems side by side corresponds to a tensor product. *Every* object in that list is a piece of linear algebra, which is why an entire lecture course on quantum algorithms can be read as applied linear algebra over $\mathbb{C}$.

Two features of the complex field matter. First, $\mathbb{C}$ is algebraically closed, so every operator has eigenvalues; the spectral theorem of Block 3 would fail over $\mathbb{R}$. Second, relative complex phases are physically observable through interference, the mechanism every quantum algorithm exploits, while *global* phases are not, which is a redundancy that is removed by the Bloch sphere of Block 2.

1.1 Vector Spaces

A *vector space* V over the field of complex numbers $\mathbb{C}$ is a non-empty set of elements called vectors. In V , it is defined the operations of vector addition and multiplication of a vector by a scalar in $\mathbb{C}$. The addition operation is associative and commutative. It also obeys properties

- There is an element $\mathbf{0} \in V$, such that, for each $\mathbf{v} \in V$, $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$ (existence of neutral element)
- For each $\mathbf{v} \in V$, there exists $\mathbf{u} = (-1)\mathbf{v}$ in V such that $\mathbf{v} + \mathbf{u} = \mathbf{0}$ (existence of inverse element)

$\mathbf{0}$ is called zero vector. The scalar multiplication operation obeys properties

- $a \cdot (b \cdot \mathbf{v}) = (a \cdot b) \cdot \mathbf{v}$ (associativity)
- $1 \cdot \mathbf{v} = \mathbf{v}$ (1 is the neutral element of multiplication)
- $(a + b) \cdot \mathbf{v} = a \cdot \mathbf{v} + b \cdot \mathbf{v}$ (distributivity of sum of scalars)
- $a \cdot (\mathbf{v} + \mathbf{w}) = a \cdot \mathbf{v} + a \cdot \mathbf{w}$ (distributivity in V)

where $\mathbf{v}, \mathbf{w} \in V$ and $a, b \in \mathbb{C}$.

A vector space can be infinite, but in most applications in *quantum computation*, *finite vector spaces* are used and are denoted by $\mathbb{C}^n$. In this case, the vectors have n complex entries. In these notes, we rarely use infinite spaces, and in these few cases, we are interested only in finite subspaces. In the context of general quantum mechanics, *infinite vector spaces* are commonly used.

A *basis* for $\mathbb{C}^n$ consists of n linearly independent vectors. If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis for $\mathbb{C}^n$, then a generic vector $\mathbf{v}$ can be written as

$$\mathbf{v} = \sum_{i=1}^n a_i \mathbf{v}_i, \quad (1)$$

where coefficients a_i are complex numbers. The *dimension* of a vector space is the number of basis vectors.

Remark 1.1. The two lists above are the axioms one finds in a first linear algebra course; the only difference from the real case is that the scalars are drawn from $\mathbb{C}$. Note also that a vector space over $\mathbb{C}$ of dimension n is a vector space over $\mathbb{R}$ of dimension $2n$: the space $\mathbb{C}^2$ of a single qubit is a four-real-dimensional object, a property we shall use when counting the parameters of a qubit in Block 2.

Example 1.2. The set $\mathbb{C}^2$ with entrywise addition and scalar multiplication is a vector space of dimension 2. The vectors

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

form a basis, and so do

$$\mathbf{f}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{f}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Bases are never unique, and a recurring theme of these lectures is that choosing the basis adapted to the operator at hand (its eigenbasis) simplifies calculations.

1.2 Inner Product

The *inner product* is a binary operation $(\cdot, \cdot) : V \times V \mapsto \mathbb{C}$, which obeys the following properties:

1. $(\cdot, \cdot)$ is linear in the second argument

$$\left(\mathbf{v}, \sum_{i=1}^n a_i \mathbf{v}_i \right) = \sum_{i=1}^n a_i (\mathbf{v}, \mathbf{v}_i).$$

2. $(\mathbf{v}_1, \mathbf{v}_2) = (\mathbf{v}_2, \mathbf{v}_1)^*$.
3. $(\mathbf{v}, \mathbf{v}) \geq 0$. The equality holds if and only if $\mathbf{v} = \mathbf{0}$.

In general, the inner product is not linear in the first argument. The property in question is called *conjugate-linear*.

There is more than one way to define an inner product on a vector space. In $\mathbb{C}^n$, the most used inner product is defined as follows: If

$$\mathbf{v} = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix},$$

then

$$(\mathbf{v}, \mathbf{w}) = \sum_{i=1}^n a_i^* b_i. \quad (2)$$

This expression is equivalent to the matrix product of the transpose-conjugate vector, which is usually denoted by $\mathbf{v}^\dagger$, by $\mathbf{w}$.

Proposition 1.3 (conjugate-linearity). *Properties 1 and 2 above force the inner product to be conjugate-linear in the first argument:*

$$\left(\sum_i a_i \mathbf{v}_i, \mathbf{w} \right) = \sum_i a_i^* (\mathbf{v}_i, \mathbf{w}).$$

Proof. Using property 2 twice and property 1 in between,

$$\left(\sum_i a_i \mathbf{v}_i, \mathbf{w} \right) = \left(\mathbf{w}, \sum_i a_i \mathbf{v}_i \right)^* = \left(\sum_i a_i (\mathbf{w}, \mathbf{v}_i) \right)^* = \sum_i a_i^* (\mathbf{w}, \mathbf{v}_i)^* = \sum_i a_i^* (\mathbf{v}_i, \mathbf{w}). \quad \square$$

Remark 1.4. Convention matters. Physicists put the conjugation on the *first* slot, as above, so that Dirac notation reads left to right; many mathematics texts conjugate the second slot instead. Everything in these notes uses the physicists' convention, but (2) is the unambiguous definition.

Example 1.5. In $\mathbb{C}^3$, take $\mathbf{v} = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} 2 \\ 0 \\ 1-i \end{bmatrix}$. Then

$$(\mathbf{v}, \mathbf{w}) = 1^* \cdot 2 + i^* \cdot 0 + 0^* \cdot (1-i) = 2, \quad (\mathbf{w}, \mathbf{v}) = 2^* \cdot 1 + 0 + (1-i)^* \cdot 0 = 2 = (\mathbf{v}, \mathbf{w})^*,$$

consistent with property 2, and $(\mathbf{v}, \mathbf{v}) = 1 + 1 + 0 = 2 \geq 0$, so $\|\mathbf{v}\| = \sqrt{2}$.

If an inner product is introduced in a vector space, we can define the notion of orthogonality. Two vectors are *orthogonal* if the inner product is zero. We also introduce the notion of *norm* using the inner product. The norm of $\mathbf{v}$, denoted by $\|\mathbf{v}\|$, is defined as

$$\|\mathbf{v}\| = \sqrt{(\mathbf{v}, \mathbf{v})}.$$

A *normalized vector* or *unit vector* is a vector whose norm is equal to 1. A basis is said *orthonormal* if all vectors are normalized and mutually orthogonal.

A finite vector space with an inner product is called a *Hilbert space*. In order for an infinite vector space to be a Hilbert space, it must obey additional properties besides having an inner product. Since we will deal primarily with finite vector spaces, we use the term *Hilbert space* as a synonym for *vector space with an inner product*. A *subspace* W of a finite Hilbert space V is also a Hilbert space. The set of vectors orthogonal to all vectors of W is the Hilbert space $W^\perp$ called *orthogonal complement*. V is the direct sum of W and $W^\perp$, that is, $V = W \oplus W^\perp$. An N -dimensional Hilbert space will be denoted by $\mathcal{H}^N$ to highlight its dimension. A Hilbert space associated with a system A will be denoted by $\mathcal{H}_A$.

Remark 1.6. The decomposition $V = W \oplus W^\perp$ is the linear-algebra skeleton of a quantum measurement with two outcomes: “the state is in W ” and “the state is in $W^\perp$ ”. In Block 3 the same decomposition reappears as a pair of complementary projectors P and $I - P$.

1.3 The Dirac Notation

In this review of linear algebra, we will systematically be using the *Dirac* or *bra-ket notation*, which was introduced by the English physicist Paul Dirac in the context of quantum mechanics to aid algebraic manipulations. This notation is very simple. Several notations are used for vectors, such as $\mathbf{v}$ and $\tilde{\mathbf{v}}$. The Dirac notation uses

$$\mathbf{v} \equiv |v\rangle.$$

Up to this point, instead of using bold or putting an arrow over letter v , we put letter v between a vertical bar and a right angle bracket. If we have an indexed basis, that is, $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, in the Dirac notation we use the form $\{|v_1\rangle, \dots, |v_n\rangle\}$ or $\{|1\rangle, \dots, |n\rangle\}$. Note that if we are using a single basis, letter $\mathbf{v}$ is unnecessary in principle. Computer scientists usually start counting from 0. So, the first basis vector is usually called $\mathbf{v}_0$. In the Dirac notation we have

$$\mathbf{v}_0 \equiv |0\rangle.$$

Vector $|0\rangle$ is not the zero vector, it is only the first vector in a collection of vectors. In the Dirac notation, the zero vector is an exception, whose notation is not modified. Here we use the notation $\mathbf{0}$.

Suppose that vector $|v\rangle$ has the following entries in a basis

$$|v\rangle = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}.$$

The *dual* vector is denoted by $\langle v|$ and is defined by

$$\langle v| = [a_1^* \quad \cdots \quad a_n^*].$$

Vectors and their duals can be seen as column and row matrices, respectively, for algebraic manipulation. The matrix product of $\langle v|$ by $|v\rangle$ is denoted by $\langle v|v\rangle$ and its value in terms of their entries is

$$\langle v|v\rangle = \sum_{i=1}^n a_i^* a_i.$$

This is an example of an inner product, which can be elegantly defined via the Dirac notation. If $\{|v_1\rangle, \dots, |v_n\rangle\}$ is an orthonormal basis, then

$$\langle v_i|v_j\rangle = \delta_{ij}, \tag{3}$$

where δ_{ij} is the *Kronecker delta*, a function that has the property $\delta_{ij} = 1$ if $i = j$ and 0 otherwise. The norm of a vector in this notation is

$$\| |v\rangle \| = \sqrt{\langle v|v\rangle}.$$

We use the terminology *ket* for vector $|v\rangle$ and *bra* for dual vector $\langle v|$. Keeping consistency, we use the terminology *bra-ket* for $\langle v|v\rangle$.

It is also very common to meet the matrix product of $|v\rangle$ by $\langle v|$, denoted by $|v\rangle\langle v|$, known as the *outer product*, whose result is an $n \times n$ matrix

$$\begin{aligned} |v\rangle\langle v| &= \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \cdot [a_1^* \quad \cdots \quad a_n^*] \\ &= \begin{bmatrix} a_1 a_1^* & \cdots & a_1 a_n^* \\ \vdots & \ddots & \vdots \\ a_n a_1^* & \cdots & a_n a_n^* \end{bmatrix}. \end{aligned} \tag{4}$$

The key to the Dirac notation is to always view *kets* as column matrices, *bras* as row matrices, and recognize that a sequence of *bras* and *kets* is a matrix product, hence associative, but non-commutative.

Example 1.7 (reading a bra-ket string). Because a bra-ket string is a matrix product, one may bracket it any way one likes. For instance

$$\underbrace{|u\rangle}_{n \times 1} \underbrace{\langle v|}_{1 \times n} \underbrace{|w\rangle}_{n \times 1} = (|u\rangle \langle v|) |w\rangle = |u\rangle \langle v|w\rangle = \langle v|w\rangle |u\rangle,$$

which says: applying the operator $|u\rangle\langle v|$ to $|w\rangle$ gives the vector $|u\rangle$ scaled by the number $\langle v|w\rangle$. Sizes are the quickest sanity check available: $\langle \cdot | \cdot \rangle$ is 1×1 , a *number*, and can be moved anywhere in a product, while $|\cdot\rangle\langle\cdot|$ is $n \times n$, an *operator*, and cannot.

Proposition 1.8 (coefficients in an orthonormal basis). *If $\{|v_1\rangle, \dots, |v_n\rangle\}$ is an orthonormal basis and $|v\rangle = \sum_i a_i |v_i\rangle$, then $a_i = \langle v_i|v\rangle$ and consequently*

$$|v\rangle = \sum_{i=1}^n \langle v_i|v\rangle |v_i\rangle, \quad \langle v|v\rangle = \sum_{i=1}^n |a_i|^2.$$

Proof. Apply $\langle v_j|$ to both sides of $|v\rangle = \sum_i a_i |v_i\rangle$ and use (3): $\langle v_j|v\rangle = \sum_i a_i \delta_{ji} = a_j$. For the norm, expand $\langle v|v\rangle = \sum_{i,j} a_i^* a_j \delta_{ij}$. $\square$

Extracting a coefficient by “hitting the state with a bra” is the single most frequently used move in the whole subject; in Block 3 we will see it is nothing but the completeness relation in disguise.

1.4 Computational Basis

The *computational basis* of $\mathbb{C}^n$, denoted by $\{|0\rangle, \dots, |n-1\rangle\}$, is given by

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad |n-1\rangle = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}.$$

This basis is also known as *canonical basis*. A few times we will use the numbering of the computational basis beginning with $|1\rangle$ and ending with $|n\rangle$. In these notes, when we use a small-caption *Latin letter* within a *ket* or *bra*, we are referring to the computational basis. Then, the following expression will always be valid

$$\langle i|j\rangle = \delta_{ij}.$$

The normalized sum of all computational basis vectors defines vector

$$|D\rangle = \frac{1}{\sqrt{n}} \sum_{i=0}^{n-1} |i\rangle, \tag{5}$$

which we will call *diagonal state*. When $n = 2$, the diagonal state is given by $|D\rangle = |+\rangle$ where

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}.$$

Remark 1.9. The diagonal state is the standard starting point of essentially every quantum algorithm: a uniform superposition over all n inputs, prepared in Block 4 by a single layer of Hadamard gates. Note that $|D\rangle$ has n nonzero amplitudes but is described by one line of notation; this economy of description is where the potential advantage of quantum computation begins, and also where the difficulty of extracting an answer begins, since a measurement of $|D\rangle$ returns a uniformly random i and nothing more.

Exercises for this block: [Exercise A.1](#) (outer products in $\mathbb{C}^3$); supplementary [Exercise S.1](#) (checking vector-space axioms), [Exercise S.2](#) (inner products and norms), [Exercise S.3](#) (bra-ket gymnastics).

2 Block 2: The Qubit, the Bloch Sphere, and Linear Operators

Learning goals:

- write a general qubit state, state the normalisation constraint, and explain why a qubit has two, not four, real parameters;
- place $|0\rangle$, $|1\rangle$, $|\pm\rangle$ and $(|0\rangle \pm i|1\rangle)/\sqrt{2}$ on the Bloch sphere and convert between (α, β) and (θ, ϕ) ;
- state the Born rule and compute measurement probabilities in the computational basis;
- define a linear operator, its rank, kernel and nullity, and use the fact that an operator is determined by its action on a basis;
- build the matrix representation of an operator from its action on a basis, in both index and outer-product form, and multiply matrices to compose operators.

2.1 Qubit and the Bloch Sphere

The *qubit* is a *unit vector* in vector space $\mathbb{C}^2$. A generic qubit $|\psi\rangle$ is represented by

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad (6)$$

where coefficients α and β are complex numbers and obey the constraint

$$|\alpha|^2 + |\beta|^2 = 1. \quad (7)$$

The set $\{|0\rangle, |1\rangle\}$ is the computational basis of $\mathbb{C}^2$ and α, β are called amplitudes of state $|\psi\rangle$. The term *state* (or *state vector*) is used as a synonym for *unit vector in a Hilbert space*.

In principle, we need four real numbers to describe a qubit, two for α and two for β . The constraint $|\alpha|^2 + |\beta|^2 = 1$ reduces to three numbers. In quantum mechanics, two vectors that differ from a *global phase factor* are considered equivalent. A global phase factor is a complex number of unit modulus multiplying the state. By eliminating this factor, a qubit can be described by two real numbers θ and ϕ as follows:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle, \quad (8)$$

where $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$. In another notation, state $|\psi\rangle$ can be represented by a point on the surface of a sphere of unit radius, called *Bloch sphere*. Numbers θ and ϕ are spherical

angles that locate the point that describes $|\psi\rangle$, as shown in Fig. 1. The vector showed there is given by

$$\begin{bmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{bmatrix}.$$

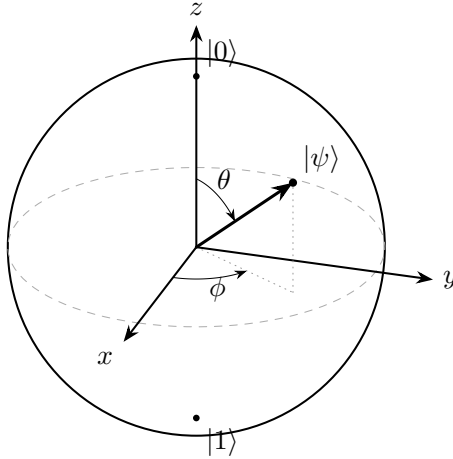


Figure 1: Bloch Sphere. The state $|\psi\rangle$ of a qubit is represented by a point on the sphere. The angles θ and ϕ are the spherical coordinates of that point, related to the amplitudes by (8).

When we disregard global phase factors, there is a one-to-one correspondence between the quantum states of a qubit and the points on the Bloch sphere. State $|0\rangle$ is in the *north pole* of the sphere, because it is obtained by taking $\theta = 0$. State $|1\rangle$ is in the *south pole*. States

$$|\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}$$

are the intersection points of the x -axis and the sphere, and states $(|0\rangle \pm i|1\rangle)/\sqrt{2}$ are the intersection points of the y -axis with the sphere.

The representation of *classical bits* in this context is given by the poles of the Bloch sphere and the representation of the *probabilistic classical bit*, that is, 0 with probability p and 1 with probability $1 - p$, is given by the point in z -axis with coordinate $2p - 1$. The interior of the Bloch sphere is used to describe the states of a qubit in the presence of *decoherence*.

Remark 2.1 (On a potential polar misunderstanding). A point on the sphere is a *superposition*, not a hidden classical alternative. The states $|+\rangle$ and $|-\rangle$ have identical measurement statistics in the computational basis, both give 0 and 1 with probability $1/2$, yet they sit at antipodal points and are perfectly distinguishable by a measurement in the $\{|+\rangle, |-\rangle\}$ basis. The relative phase between α and β , invisible in the computational basis alone, is the angle ϕ .

Pro Memoria: measurement and the Born rule

Linear algebra alone does not say what a measurement is; one physical postulate is needed. Measuring $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ in the computational basis returns the outcome $i \in \{0, 1\}$ with probability

$$p(i) = |\langle i|\psi\rangle|^2 = \langle\psi|i\rangle\langle i|\psi\rangle,$$

and leaves the system in state $|i\rangle$. Hence $p(0) = |\alpha|^2$, $p(1) = |\beta|^2$, and the normalisation (7) is equivalent to the statement $p(0) + p(1) = 1$. More generally, for any orthonormal basis $\{|v_i\rangle\}$ the outcome probabilities are $p(i) = |\langle v_i|\psi\rangle|^2$, which is why the coefficient-extraction

proposition of Block 1 is a physically meaningful operation and not just bookkeeping. Two observations follow immediately:

- A global phase $e^{i\gamma} |\psi\rangle$ changes no probability $|\langle v_i | \psi \rangle|^2$ in any basis, this is why global phases are discarded and why the Bloch sphere is a useful visualization.
- Relative phases *do* change probabilities in other bases; the operator $P_i = |i\rangle\langle i|$ used above is the projector introduced in Block 3.

2.2 Linear Operators

Let V and W be vector spaces, $\{|v_1\rangle, \dots, |v_n\rangle\}$ a basis for V , and $\mathcal{A}$ a function $\mathcal{A} : V \mapsto W$ that satisfies

$$\mathcal{A}\left(\sum_i a_i |v_i\rangle\right) = \sum_i a_i \mathcal{A}(|v_i\rangle), \quad (9)$$

for any complex numbers a_i . $\mathcal{A}$ is called a *linear operator* from V to W . The term *linear operator in V* means that both the domain and codomain of $\mathcal{A}$ is V . The composition of linear operators $\mathcal{A} : V_1 \mapsto V_2$ and $\mathcal{B} : V_2 \mapsto V_3$ is also a linear operator $\mathcal{C} : V_1 \mapsto V_3$ obtained through the composition of their functions: $\mathcal{C}(|v\rangle) = \mathcal{B}(\mathcal{A}(|v\rangle))$. The sum of two linear operators, both from V to W , is defined by the formula $(\mathcal{A} + \mathcal{B})(|v\rangle) = \mathcal{A}(|v\rangle) + \mathcal{B}(|v\rangle)$.

The *identity* operator $\mathcal{I}$ in V is a linear operator such that $\mathcal{I}(|v\rangle) = |v\rangle$ for all $|v\rangle \in V$. The *null* operator $\mathcal{O}$ in V is a linear operator such that $\mathcal{O}(|v\rangle) = \mathbf{0}$ for all $|v\rangle \in V$.

The *rank* of a linear operator $\mathcal{A}$ in V is the dimension of the image of $\mathcal{A}$. The *kernel* or *nullspace* of a linear operator $\mathcal{A}$ in V is the set of all vectors $|v\rangle$ for which $\mathcal{A}(|v\rangle) = \mathbf{0}$. The dimension of the kernel is called the *nullity* of the operator. The *rank-nullity theorem* states that $\text{rank } \mathcal{A} + \text{nullity } \mathcal{A} = \dim V$.

Fact 2.2. *If we specify the action of a linear operator on a basis of vector space V , its action on any vector in V can be straightforwardly determined.*

Reason: Every $|v\rangle \in V$ has a unique expansion $|v\rangle = \sum_i a_i |v_i\rangle$ in the chosen basis, and (9) then forces $\mathcal{A}|v\rangle = \sum_i a_i \mathcal{A}|v_i\rangle$. Thus the n vectors $\mathcal{A}|v_i\rangle$ determine $\mathcal{A}$ completely, and conversely any choice of those n vectors defines a linear operator. This is the reason that specifying a quantum gate on the computational basis, one column at a time — specifies it everywhere, and the reason that operators can be stored as matrices at all. $\square$

Remark 2.3. The kernel and the rank-nullity theorem are less prominent in quantum computation than elsewhere in linear algebra, for a good reason: the operators that implement quantum gates are unitary (Block 3), hence invertible, so their kernel is $\{\mathbf{0}\}$ and their rank is full. Non-invertible operators do appear, but as *projectors* describing measurements, where the kernel is the subspace of states that outcome rules out.

2.3 Matrix Representation

Linear operators can be represented by matrices. Let $\mathcal{A} : V \mapsto W$ be a linear operator, $\{|v_1\rangle, \dots, |v_n\rangle\}$ and $\{|w_1\rangle, \dots, |w_m\rangle\}$ orthonormal bases for V and W , respectively. A *matrix representation* of $\mathcal{A}$ is obtained by applying $\mathcal{A}$ to every vector in the basis of V and expressing the result as a linear combination of basis vectors of W , as follows:

$$\mathcal{A}(|v_j\rangle) = \sum_{i=1}^m a_{ij} |w_i\rangle, \quad (10)$$

where index j run from 1 to n . Therefore, a_{ij} are entries of an $m \times n$ matrix, which we call A . In this case, expression $\mathcal{A}(|v_j\rangle)$, which means function $\mathcal{A}$ applied to argument $|v_j\rangle$, is equivalent to the matrix product $A|v_j\rangle$. Using the outer product notation, we have

$$A = \sum_{i=1}^m \sum_{j=1}^n a_{ij} |w_i\rangle\langle v_j|. \quad (11)$$

Using the above equation and the property that the basis of V is orthonormal, we can verify that the matrix product of A by $|v_j\rangle$ is equal to $\mathcal{A}(|v_j\rangle)$. The key to this calculation is to use the associativity of matrix multiplication:

$$\begin{aligned} (|w_i\rangle\langle v_j|) |v_k\rangle &= |w_i\rangle (\langle v_j|v_k\rangle) \\ &= \delta_{jk} |w_i\rangle. \end{aligned}$$

In particular, the matrix representation of the identity operator $\mathcal{I}$ in any orthonormal basis is the identity matrix I and the matrix representation of the null operator $\mathcal{O}$ in any orthonormal basis is the *zero matrix*.

If the linear operator $\mathcal{C}$ is the composition of the linear operators $\mathcal{B}$ and $\mathcal{A}$, the matrix representation of $\mathcal{C}$ will be obtained by multiplying the matrix representation of $\mathcal{B}$ with that of $\mathcal{A}$, that is, $C = BA$.

When we fix orthonormal bases for the vector spaces, there is a one-to-one correspondence between linear operators and matrices. In $\mathbb{C}^n$, we use the computational basis as a reference basis, so the terms *linear operator* and *matrix* are taken as synonyms. We will also use the term *operator* as a synonym for *linear operator*.

Proposition 2.4 (entries are matrix elements). *With the notation above, $a_{ij} = \langle w_i|A|v_j\rangle$; and if $C = BA$ then $c_{ik} = \sum_j b_{ij}a_{jk}$.*

Proof. Apply $\langle w_k|$ to (10) and use orthonormality: $\langle w_k|A|v_j\rangle = \sum_i a_{ij} \langle w_k|w_i\rangle = a_{kj}$. For the product, insert the definition twice:

$$\mathcal{B}\mathcal{A}|v_k\rangle = \mathcal{B}\left(\sum_j a_{jk} |u_j\rangle\right) = \sum_j a_{jk} \sum_i b_{ij} |w_i\rangle = \sum_i \left(\sum_j b_{ij}a_{jk}\right) |w_i\rangle,$$

which is the familiar row-times-column rule. Matrix multiplication is thus not a definition to be memorised but a consequence of composing linear maps. $\square$

Example 2.5 (three ways to write one gate). Consider the operator on $\mathbb{C}^2$ defined on the computational basis by $X|0\rangle = |1\rangle$ and $X|1\rangle = |0\rangle$. By the property above this determines X everywhere. Its three equivalent descriptions are

$$\text{action: } X|j\rangle = |1-j\rangle, \quad \text{outer product: } X = |0\rangle\langle 1| + |1\rangle\langle 0|, \quad \text{matrix: } X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Reading (11) backwards is often the fastest route: the coefficient of $|i\rangle\langle j|$ is the entry in row i , column j . As a check, $X|+\rangle = |+\rangle$ and $X|-\rangle = -|-\rangle$, so $|\pm\rangle$ are eigenvectors of X with eigenvalues ± 1 , the diagonal representation of Block 3.

Example 2.6 (columns are images of basis vectors). The Hadamard operator is defined by $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$. Placing those two images as columns gives

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

and a direct calculation gives $H^2 = I$, so H is its own inverse. Applied to $|0\rangle$ it produces the diagonal state $|D\rangle$ of (5) for $n = 2$; [Exercise A.15](#) generalises this to n qubits.

Pro Memoria: gates are matrices, circuits are matrix products

A quantum gate is a linear operator on the state space of the qubits it touches, and by the property above it is fully specified by its action on the computational basis. A circuit that applies G_1 , then G_2 , then G_3 implements the single operator $G_3G_2G_1$, note the reversal: circuit diagrams are read left to right, matrix products right to left. Two consequences we will need later: the operator implemented by a circuit of ℓ gates is a product of ℓ matrices, and since each gate will turn out to be unitary (Section 3.4), so is the whole circuit; and the cost of a quantum algorithm is counted in *gates*, not in the 4^n entries of the matrix they multiply out to.

Exercises for this block: [Exercise A.2](#), [Exercise A.3](#), [Exercise A.4](#) (Bloch sphere); [Exercise A.5](#) (matrix representation and the Hadamard operator); supplementary [Exercise S.4](#) (Born rule practice), [Exercise S.5](#) (Bloch coordinates), [Exercise S.6](#) (operators from their action on a basis).

Part II

Lecture 2, Spectra, Special Operators, Tensor Products

This second lecture attempts to introduce the tools that are used in quantum computing. The first is *diagonalisation*: writing an operator as a weighted sum of projectors, which turns operator functions, time evolution and measurement into manipulations of numbers. The second is the *tensor product*, which is how several qubits are assembled into a register and the reason the state space grows exponentially with the number of qubits.

3 Block 3: Diagonal Representation, Completeness, Special Operators

Learning goals:

- write a diagonalisable operator as $\sum_i \lambda_i |v_i\rangle\langle v_i|$ and as $\sum_\lambda \lambda P_\lambda$, and construct the projectors P_λ from an eigenbasis;
- use the completeness relation $I = \sum_i |v_i\rangle\langle v_i|$ to insert a basis anywhere in a calculation;
- state and use the Cauchy–Schwarz inequality;
- compute adjoints and manipulate the dagger through products, sums and outer products;
- decide whether a given operator is normal, unitary, Hermitian, a projector, or positive, and recite the consequences of each;
- write down the Pauli matrices, their algebraic properties and their action on the computational basis.

3.1 Diagonal Representation

Let $\mathcal{O}$ be an operator in V . If there exists an orthonormal basis $\{|v_1\rangle, \dots, |v_n\rangle\}$ of V such that

$$\mathcal{O} = \sum_{i=1}^n \lambda_i |v_i\rangle\langle v_i|, \quad (12)$$

we say that $\mathcal{O}$ admits a *diagonal representation* or, equivalently, $\mathcal{O}$ is *diagonalizable*. The complex numbers λ_i are the *eigenvalues* of $\mathcal{O}$ and $|v_i\rangle$ are the corresponding *eigenvectors*. Any multiple of an eigenvector is also an eigenvector. If two eigenvectors are associated with the same eigenvalue, then any linear combination of these eigenvectors is an eigenvector. The number of linearly independent eigenvectors associated with the same eigenvalue is the *multiplicity* of that eigenvalue.

If there are eigenvalues with multiplicity greater than one, the diagonal representation can be factored out as follows:

$$\mathcal{O} = \sum_{\lambda} \lambda P_{\lambda}, \quad (13)$$

where index λ runs only on the distinct eigenvalues and P_{λ} is the projector on the eigenspace of $\mathcal{O}$ associated with eigenvalue λ . If λ has multiplicity 1, $P_{\lambda} = |v\rangle\langle v|$, where $|v\rangle$ is the unit

eigenvector associated with λ . If λ has multiplicity 2 and $|v_1\rangle, |v_2\rangle$ are linearly independent unit eigenvectors associated with λ , $P_\lambda = |v_1\rangle\langle v_1| + |v_2\rangle\langle v_2|$ and so on. The projectors P_λ satisfy

$$\sum_{\lambda} P_{\lambda} = I. \quad (14)$$

An alternative way to define a diagonalizable operator is by requiring that O is *similar* to a diagonal matrix. Matrices O and O' are similar if $O' = M^{-1}OM$ for some invertible matrix M . We have interest only in the case when M is a unitary matrix. The term *diagonalizable* we use here is narrower than the one used in the literature, because we are demanding that M be a unitary matrix.

Remark 3.1 (Using (12) as a recipe). Equation (12) says: to apply O to a state, decompose the state into eigenvectors, multiply each piece by its eigenvalue, and add up. Indeed

$$O|\psi\rangle = \sum_i \lambda_i |v_i\rangle \langle v_i|\psi\rangle,$$

so in the eigenbasis O acts as multiplication by numbers. Everything in Block 4 about operator functions (square roots, exponentials, t -th powers) is this equation applied to $f(\lambda_i)$ instead of λ_i .

Example 3.2. The Pauli operator X of Section 3.5 satisfies $X|\pm\rangle = \pm|\pm\rangle$, so its diagonal representation is

$$X = (+1)|+\rangle\langle +| + (-1)|-\rangle\langle -| = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \checkmark$$

Both eigenvalues have multiplicity 1, and $P_{+1} + P_{-1} = |+\rangle\langle +| + |-\rangle\langle -| = I$, illustrating (14). The identity I itself is the opposite: a single eigenvalue 1 of multiplicity n , so $P_1 = I$ and any orthonormal basis is an eigenbasis.

3.2 Completeness Relation

The *completeness relation* is so useful that it deserves to be highlighted. Let $\{|v_1\rangle, \dots, |v_n\rangle\}$ be an orthonormal basis of V . Then,

$$I = \sum_{i=1}^n |v_i\rangle\langle v_i|. \quad (15)$$

The completeness relation is the diagonal representation of the identity matrix.

Proof. Both sides are linear operators, so by what we learned in Block 2 it suffices to check them on a basis. Applying the right-hand side to $|v_j\rangle$ gives $\sum_i |v_i\rangle\langle v_i|v_j\rangle = \sum_i \delta_{ij} |v_i\rangle = |v_j\rangle$, i.e. the Identity operation. $\square$

How to use (15). You can insert $I = \sum_i |v_i\rangle\langle v_i|$ and this creates a sum over a convenient basis. For example take the expansion of a state or the definition of matrix entries, and the trace in Dirac form appear from one insertion of (15):

$$|\psi\rangle = I|\psi\rangle = \sum_i |v_i\rangle \langle v_i|\psi\rangle, \quad A = IAI = \sum_{i,j} |v_i\rangle \langle v_i|A|v_j\rangle \langle v_j|.$$

The second identity is (11) that you will see in Block 4 and is used in Exercise A.7.

3.3 Cauchy–Schwarz Inequality

Let V be a Hilbert space and $|v\rangle, |w\rangle \in V$. Then,

$$|\langle v|w\rangle| \leq \sqrt{\langle v|v\rangle \langle w|w\rangle}. \quad (16)$$

A more explicit way of presenting the Cauchy–Schwarz inequality is

$$\left| \sum_i v_i w_i \right|^2 \leq \left(\sum_i |v_i|^2 \right) \left(\sum_i |w_i|^2 \right),$$

which is obtained when we take $|v\rangle = \sum_i v_i^* |i\rangle$ and $|w\rangle = \sum_i w_i |i\rangle$.

Proof. If $|w\rangle = \mathbf{0}$ both sides vanish. Otherwise put $|u\rangle = |v\rangle - \frac{\langle w|v\rangle}{\langle w|w\rangle} |w\rangle$, which satisfies $\langle w|u\rangle = 0$. Then

$$0 \leq \langle u|u\rangle = \langle v|v\rangle - \frac{|\langle w|v\rangle|^2}{\langle w|w\rangle},$$

and multiplying by $\langle w|w\rangle > 0$ gives (16). Equality holds precisely when $|u\rangle = \mathbf{0}$, that is, when $|v\rangle$ and $|w\rangle$ are proportional. $\square$

Remark 3.3. In quantum computation (16) is the reason $|\langle v|\psi\rangle|^2 \leq 1$ for unit vectors, and why the Born rule introduced in Block 2 returns probabilities; the quantity $|\langle v|\psi\rangle|$ is called the *overlap* or *fidelity* of the two states, and equals 1 when they are identical up to a global phase.

3.4 Special Operators

Let A be a linear operator in Hilbert space V . Then, there exists a unique linear operator $A^\dagger$ in V , called *adjoint operator*, that satisfies

$$(|v\rangle, A|w\rangle) = (A^\dagger|v\rangle, |w\rangle), \quad (17)$$

for all $|v\rangle, |w\rangle \in V$.

The matrix representation of $A^\dagger$ is the transpose-conjugate matrix $(A^*)^T$. The main properties of the *dagger* or *transpose-conjugate* operation are

1. $(AB)^\dagger = B^\dagger A^\dagger$
2. $|v\rangle^\dagger = \langle v|$
3. $(A|v\rangle)^\dagger = \langle v| A^\dagger$
4. $(|w\rangle \langle v|)^\dagger = |v\rangle \langle w|$
5. $(A^\dagger)^\dagger = A$
6. $(\sum_i a_i A_i)^\dagger = \sum_i a_i^* A_i^\dagger$

The last property shows that the dagger operation is *conjugate-linear* when applied on a linear combination of operators.

Practical rule. The 'dagger' an expression requires that you reverse the order of all factors and dagger each one, turning kets into bras, bras into kets, and scalars into their conjugates. Thus $(a A |u\rangle \langle v| B)^\dagger = a^* B^\dagger |v\rangle \langle u| A^\dagger$.

Normal Operator

An operator A in V is *normal* if $A^\dagger A = AA^\dagger$.

Spectral Theorem

An operator A in V is diagonalizable if and only if A is normal.

Proof sketch . If $A = \sum_i \lambda_i |v_i\rangle\langle v_i|$ with $\{|v_i\rangle\}$ orthonormal, then $A^\dagger = \sum_i \lambda_i^* |v_i\rangle\langle v_i|$ by properties 4 and 6 above, and using $|v_i\rangle\langle v_i| |v_j\rangle\langle v_j| = \delta_{ij} |v_i\rangle\langle v_i|$ one finds

$$A^\dagger A = \sum_i |\lambda_i|^2 |v_i\rangle\langle v_i| = AA^\dagger.$$

Since $\mathbb{C}$ is algebraically closed, A has an eigenvalue λ with unit eigenvector $|v\rangle$. Normality implies that the orthogonal complement of $|v\rangle$ is invariant under both A and $A^\dagger$, so A restricted to that $(n-1)$ -dimensional space is again normal; induction on the dimension produces an orthonormal eigenbasis. $\square$

Remark 3.4. The spectral theorem is the reason the notion “normal” is named: it converts an algebraic condition that can be checked by two matrix multiplications into the geometric statement that an orthonormal eigenbasis exists. [Exercise A.8](#) and [Exercise A.9](#) explore what goes wrong without it.

Unitary Operator

An operator U in V is *unitary* if $U^\dagger U = UU^\dagger = I$.

Properties Of Unitary Operators

Unitary operators are normal, so they are diagonalizable with respect to an orthonormal basis. Eigenvectors of a unitary operator associated with different eigenvalues are orthogonal. The eigenvalues have unit modulus, that is, their form is $e^{i\alpha}$, where α is a real number. Unitary operators preserve the inner product, that is, the inner product of $U|v_1\rangle$ by $U|v_2\rangle$ is equal to the inner product of $|v_1\rangle$ by $|v_2\rangle$. The application of a unitary operator on a vector preserves its norm.

. Normality comes from $U^\dagger U = I = UU^\dagger$. For the inner product, use (17):

$$\langle Uv_1 | Uv_2 \rangle = \left\langle v_1 \left| U^\dagger U \right| v_2 \right\rangle = \langle v_1 | I | v_2 \rangle = \langle v_1 | v_2 \rangle,$$

and taking $|v_1\rangle = |v_2\rangle$ gives norm preservation. If $U|v\rangle = \lambda|v\rangle$ with $\| |v\rangle \| = 1$, then $1 = \langle v | v \rangle = \langle Uv | Uv \rangle = |\lambda|^2$, so $|\lambda| = 1$ and $\lambda = e^{i\alpha}$. Finally, if $U|v_1\rangle = \lambda_1|v_1\rangle$ and $U|v_2\rangle = \lambda_2|v_2\rangle$ then $\langle v_1 | v_2 \rangle = \langle Uv_1 | Uv_2 \rangle = \lambda_1^* \lambda_2 \langle v_1 | v_2 \rangle$; since $\lambda_1^* \lambda_2 \neq 1$ when $\lambda_1 \neq \lambda_2$ on the unit circle, $\langle v_1 | v_2 \rangle = 0$. $\square$

Pro Memoria: unitarity is reversibility

Every gate of an ideal quantum computer is a unitary operator, and the properties above say why this is the right class. Norm preservation keeps $\sum_i |a_i|^2 = 1$, so probabilities stay probabilities. Inner-product preservation says that two states which were distinguishable remain equally distinguishable, and two orthogonal states stay orthogonal: no gate can increase our ability to tell states apart. And $U^\dagger U = I$ means every gate has an inverse,

namely $U^\dagger$, so quantum computation is *reversible*; irreversibility enters only through measurement, which is described by projectors rather than unitaries. This is also why the reversible construction $U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$ of Exercise A.16 is needed to compute a general function f on a quantum computer.

Hermitian Operator

An operator A in V is *Hermitian* or *self-adjoint* if $A^\dagger = A$.

Properties of Hermitian Operators

Hermitian operators are normal, so they are diagonalizable with respect to an orthonormal basis. Eigenvectors of a Hermitian operator associated with different eigenvalues are orthogonal. The eigenvalues of a Hermitian operator are real numbers. A real symmetric matrix is Hermitian.

Proofs of the properties. If $A|v\rangle = \lambda|v\rangle$ with $\| |v\rangle \| = 1$, then $\lambda = \langle v|A|v\rangle = \langle v|A^\dagger|v\rangle = \langle v|A|v\rangle^* = \lambda^*$, so $\lambda \in \mathbb{R}$. For orthogonality, let $A|v_1\rangle = \lambda_1|v_1\rangle$ and $A|v_2\rangle = \lambda_2|v_2\rangle$ with $\lambda_1 \neq \lambda_2$. Computing $\langle v_1|A|v_2\rangle$ in two ways gives $\lambda_2 \langle v_1|v_2\rangle = \lambda_1 \langle v_1|v_2\rangle$ (using $\lambda_1^* = \lambda_1$), hence $(\lambda_1 - \lambda_2) \langle v_1|v_2\rangle = 0$ and $\langle v_1|v_2\rangle = 0$. $\square$

Remark 3.5. Physical observables are represented by Hermitian operators because their eigenvalues, the possible measurement outcomes, must be real, and their eigenvectors form an orthonormal basis, the mutually exclusive outcomes. The Pauli matrices of Section 3.5 are simultaneously Hermitian and unitary, so they are both observables and gates; an operator is both if and only if $A^2 = I$.

Orthogonal Projector

An operator P in V is an *orthogonal projector* if $P^2 = P$ and $P^\dagger = P$.

Properties of Orthogonal Projectors

The eigenvalues are equal to 0 or 1. If P is an orthogonal projector, then the *orthogonal complement* $I - P$ is also an orthogonal projector. Applying a projector on a vector either decreases its norm or maintains invariant. In these notes, we use the term *projector* as a synonym for *orthogonal projector*. We will use the term *non-orthogonal projector* explicitly to distinguish this case. An example of a non-orthogonal projector on a qubit is $P = |1\rangle\langle +|$.

Proofs of the properties. If $P|v\rangle = \lambda|v\rangle$ then $\lambda|v\rangle = P|v\rangle = P^2|v\rangle = \lambda^2|v\rangle$, so $\lambda^2 = \lambda$ and $\lambda \in \{0, 1\}$. For the complement, $(I - P)^2 = I - 2P + P^2 = I - P$ and $(I - P)^\dagger = I - P$. For the norm, write $|v\rangle = P|v\rangle + (I - P)|v\rangle$; the two pieces are orthogonal, since $(P|v\rangle, (I - P)|v\rangle) = \langle v|P(I - P)|v\rangle = 0$, so $\| |v\rangle \|^2 = \| P|v\rangle \|^2 + \| (I - P)|v\rangle \|^2 \geq \| P|v\rangle \|^2$. $\square$

Pro Memoria: projectors are measurements

A projective measurement is a set of projectors $\{P_k\}$ with $\sum_k P_k = I$, precisely (14). Outcome k occurs with probability $p(k) = \langle \psi|P_k|\psi\rangle = \|P_k|\psi\rangle\|^2$, and the post-measurement state is $P_k|\psi\rangle / \sqrt{p(k)}$: the projector removes the part of the state inconsistent with the outcome, and the norm decrease proved above is the probability of that outcome. Measuring an observable $A = \sum_\lambda \lambda P_\lambda$ means performing this measurement and reporting the eigenvalue λ ; the average result over many repetitions is $\sum_\lambda \lambda p(\lambda) = \langle \psi|A|\psi\rangle$, the

expectation value. The measurement of Block 2 is the special case $P_i = |i\rangle\langle i|$.

Positive Operator

An operator A in V is said *positive* if $\langle v|A|v\rangle \geq 0$ for any $|v\rangle \in V$. If the inequality is strict for any nonzero vector in V , then the operator is said *positive definite*.

Properties of Positive Operators

Positive operators are Hermitian. The eigenvalues are nonnegative real numbers.

Proof of the second property. Positivity gives $\langle v|A|v\rangle \geq 0$ for all $|v\rangle$; taking $|v\rangle$ to be a unit eigenvector with eigenvalue λ yields $\lambda \geq 0$. Conversely, a Hermitian operator with nonnegative eigenvalues is positive, since $\langle v|A|v\rangle = \sum_i \lambda_i |\langle v_i|v\rangle|^2 \geq 0$ in its eigenbasis. [Exercise A.12](#) gives the standard source of positive operators. $\square$

3.5 Pauli Matrices

The *Pauli matrices* are

$$\begin{aligned}\sigma_0 &= I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \\ \sigma_1 &= \sigma_x = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \\ \sigma_2 &= \sigma_y = Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \\ \sigma_3 &= \sigma_z = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.\end{aligned}$$

These matrices are unitary and Hermitian, and hence their eigenvalues are equal to ± 1 . Putting in another way: $\sigma_j^2 = I$ and $\sigma_j^\dagger = \sigma_j$ for $j = 0, \dots, 3$.

The following properties are extensively used:

$$\begin{aligned}X|0\rangle &= |1\rangle, & X|1\rangle &= |0\rangle, \\ Z|0\rangle &= |0\rangle, & Z|1\rangle &= -|1\rangle.\end{aligned}$$

Pauli matrices form a basis for the vector space of 2×2 matrices. Therefore, a generic operator that acts on a qubit can be written as a linear combination of Pauli matrices.

Remark 3.6 (algebra). Beyond $\sigma_j^2 = I$, the Pauli matrices *anticommute* in pairs and multiply cyclically:

$$\sigma_j \sigma_k = -\sigma_k \sigma_j \quad (j \neq k, j, k \geq 1), \quad XY = iZ, \quad YZ = iX, \quad ZX = iY.$$

Because $\{I, X, Y, Z\}$ is a basis of the 2×2 matrices, any operator on a qubit is $A = c_0 I + c_1 X + c_2 Y + c_3 Z$ with $c_j = \frac{1}{2} \text{tr}(\sigma_j A)$, a formula that follows from $\text{tr}(\sigma_j \sigma_k) = 2\delta_{jk}$ and the trace properties of Block 4. If A is Hermitian, all four c_j are real.

Example 3.7 (Pauli operators as rotations). Each Pauli matrix is Hermitian with eigenvalues ± 1 , so by (13) it is the difference of two complementary projectors:

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|, \quad X = |+\rangle\langle +| - |-\rangle\langle -|, \quad Y = |+\mathrm{i}\rangle\langle +\mathrm{i}| - |-\mathrm{i}\rangle\langle -\mathrm{i}|,$$

where $|\pm i\rangle = (|0\rangle \pm i|1\rangle)/\sqrt{2}$. Geometrically each acts on the Bloch sphere as a rotation by π about the corresponding axis, which is the content of [Exercise A.13](#).

Exercises for this block: [Exercise A.6](#) (projectors from eigenvalues ± 1), [Exercise A.7](#) (completeness in reverse), [Exercise A.8](#), [Exercise A.9](#) (a non-normal and a non-unitary example), [Exercise A.10](#), [Exercise A.11](#) (closure properties), [Exercise A.12](#) (positivity of $A^\dagger A$), [Exercise A.13](#) (Pauli matrices on the Bloch sphere); supplementary [Exercise S.7](#) (dagger practice), [Exercise S.8](#) (diagonalising a Hermitian qubit operator), [Exercise S.9](#) (Pauli algebra).

4 Block 4: Operator Functions, Tensor Product, Registers

Learning goals:

- evaluate $f(A)$ for a normal operator A , in particular $\sqrt{A}$, e^A and U^t , via the diagonal representation;
- compute traces, use linearity and the cyclic property, and evaluate $\text{tr}(A|\psi\rangle\langle\psi|)$;
- form tensor products of vectors and of operators, compute them explicitly as Kronecker products, and use bilinearity and the mixed-product property;
- write the state of an n -qubit register in binary and decimal notation and count its amplitudes;
- recognise a product state and distinguish it from an entangled state.

4.1 Operator Functions

If we have an operator A in V , we can ask whether it is possible to calculate $\sqrt{A}$, that is, to find an operator the square of which is A ? In general, we can ask ourselves whether it makes sense to use an operator as an argument of a usual function, such as, exponential or logarithmic function. If operator A is normal, it has a diagonal representation, that is, can be written in the form

$$A = \sum_i a_i |v_i\rangle\langle v_i|,$$

where a_i are the eigenvalues and the set $\{|v_i\rangle\}$ is an orthonormal basis of eigenvectors of A . We can extend the application of a function $f : \mathbb{C} \mapsto \mathbb{C}$ to A as follows

$$f(A) = \sum_i f(a_i) |v_i\rangle\langle v_i|. \quad (18)$$

The result is an operator defined in the same vector space V and it is independent of the choice of basis of V .

If the goal is to calculate $\sqrt{A}$, first A must be diagonalized, that is, we must determine a unitary matrix U such that $A = UDU^\dagger$, where D is a diagonal matrix. Then, we use the property that $\sqrt{A} = U\sqrt{D}U^\dagger$, where $\sqrt{D}$ is calculated by taking the square root of each diagonal element.

If U is the evolution operator of an isolated quantum system that is initially in state $|\psi(0)\rangle$, the state at time t is given by

$$|\psi(t)\rangle = U^t |\psi(0)\rangle. \quad (19)$$

The most efficient way to calculate state $|\psi(t)\rangle$ is to obtain the diagonal representation of the unitary operator U

$$U = \sum_i \lambda_i |v_i\rangle\langle v_i|,$$

and to calculate the t -th power U , that is,

$$U^t = \sum_i \lambda_i^t |v_i\rangle\langle v_i|.$$

The system state at time t will be

$$|\psi(t)\rangle = \sum_i \lambda_i^t \langle v_i | \psi(0) \rangle |v_i\rangle. \quad (20)$$

Remark 4.1 (why (18) is basis independent). Nothing in (18) depends on which unitary U was used to diagonalise A : if $A = UDU^\dagger$ then, because $(UDU^\dagger)^k = UD^kU^\dagger$, every polynomial satisfies $p(A) = Up(D)U^\dagger$, and (18) is the limit of that statement for a general f . Two warnings, both easy to forget. First, $f(A)$ is *not* f applied entrywise to the matrix of A . Second, square roots are not unique: each eigenvalue offers two choices of sign, so a 2×2 operator with distinct nonzero eigenvalues has four square roots, which is the point of [Exercise A.14](#).

Example 4.2. Take $A = X$, with $X = (+1)|+\rangle\langle+| + (-1)|-\rangle\langle-|$. Then

$$\begin{aligned} \sqrt{X} &= |+\rangle\langle+| + i|-\rangle\langle-| = \frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}, \\ e^{-i\theta X} &= e^{-i\theta}|+\rangle\langle+| + e^{i\theta}|-\rangle\langle-| = \cos\theta I - i\sin\theta X, \end{aligned}$$

choosing the branch $\sqrt{-1} = i$. The second identity is the standard single-qubit rotation gate about the x -axis, and the same computation with Z gives the phase gate. This is the practical route to every parametrised gate: diagonalise, exponentiate the numbers, reassemble.

The trace

The *trace* of a matrix is another type of operator function. In this case, the result of applying the trace function is a complex number defined as

$$\text{tr}(A) = \sum_i a_{ii}, \quad (21)$$

where a_{ii} are the diagonal elements of A . In the Dirac notation

$$\text{tr}(A) = \sum_i \langle v_i | A | v_i \rangle, \quad (22)$$

where $\{|v_1\rangle, \dots, |v_n\rangle\}$ is an orthonormal basis of V . The trace function satisfies the following properties:

1. $\text{tr}(aA + bB) = a\text{tr}(A) + b\text{tr}(B)$, (linearity)
2. $\text{tr}(AB) = \text{tr}(BA)$,
3. $\text{tr}(ABC) = \text{tr}(CAB)$. (cyclic property)

The third property follows from the second one. Properties 2 and 3 are valid even when A , B , and C are not square matrices.

The trace function is invariant under *similarity transformations*, that is, $\text{tr}(M^{-1}AM) = \text{tr}(A)$, where M is an invertible matrix. This implies that the trace does not depend on the basis choice for the matrix representation of A .

A useful formula involving the trace of operators is

$$\text{tr}(A|\psi\rangle\langle\psi|) = \langle\psi|A|\psi\rangle, \quad (23)$$

for any $|\psi\rangle \in V$ and any A in V . This formula can be easily proved using the cyclic property of the trace function.

Proofs. Property 2: $\text{tr}(AB) = \sum_i \sum_j a_{ij} b_{ji} = \sum_j \sum_i b_{ji} a_{ij} = \text{tr}(BA)$. Property 3 follows by applying property 2 to the pair (AB, C) . Similarity invariance follows from property 2 with $A \rightarrow M^{-1}A$ and $B \rightarrow M$. For (23), use the cyclic property with the factors A , $|\psi\rangle$ and $\langle\psi|$:

$$\text{tr}(A|\psi\rangle\langle\psi|) = \text{tr}(\langle\psi|A|\psi\rangle) = \langle\psi|A|\psi\rangle,$$

the middle expression being a 1×1 matrix, whose trace is its single entry. Combined with (13), this makes $\langle\psi|A|\psi\rangle = \sum_\lambda \lambda \text{tr}(P_\lambda |\psi\rangle\langle\psi|)$ the expectation value of the observable A . $\square$

4.2 Tensor Product

Let V and W be finite Hilbert spaces with basis $\{|v_1\rangle, \dots, |v_m\rangle\}$ and $\{|w_1\rangle, \dots, |w_n\rangle\}$, respectively. The *tensor product* of V by W , denoted by $V \otimes W$, is an mn -dimensional Hilbert space, for which set $\{|v_1\rangle \otimes |w_1\rangle, |v_1\rangle \otimes |w_2\rangle, \dots, |v_m\rangle \otimes |w_n\rangle\}$ is a basis. The tensor product of a vector in V by a vector in W , such as $|v\rangle \otimes |w\rangle$, also denoted by $|v\rangle |w\rangle$ or $|v, w\rangle$ or $|vw\rangle$, can be calculated explicitly via the Kronecker product, defined ahead. A generic vector in $V \otimes W$ is a linear combination of vectors $|v_i\rangle \otimes |w_j\rangle$, that is, if $|\psi\rangle \in V \otimes W$ then

$$|\psi\rangle = \sum_{i=1}^m \sum_{j=1}^n a_{ij} |v_i\rangle \otimes |w_j\rangle. \quad (24)$$

The tensor product is *bilinear*, that is, linear in each argument:

1. $|v\rangle \otimes (a|w_1\rangle + b|w_2\rangle) = a|v\rangle \otimes |w_1\rangle + b|v\rangle \otimes |w_2\rangle,$
2. $(a|v_1\rangle + b|v_2\rangle) \otimes |w\rangle = a|v_1\rangle \otimes |w\rangle + b|v_2\rangle \otimes |w\rangle.$

A scalar can always be factored out to the beginning of the expression:

$$a(|v\rangle \otimes |w\rangle) = (a|v\rangle) \otimes |w\rangle = |v\rangle \otimes (a|w\rangle).$$

The tensor product of a linear operator A in V by B in W , denoted by $A \otimes B$, is a linear operator in $V \otimes W$ defined by

$$(A \otimes B)(|v\rangle \otimes |w\rangle) = (A|v\rangle) \otimes (B|w\rangle). \quad (25)$$

A generic linear operator in $V \otimes W$ can be written as a linear combination of operators of the form $A \otimes B$, but an operator in $V \otimes W$ cannot be factored out in general. This definition can easily be extended to operators $A : V \mapsto V'$ and $B : W \mapsto W'$. In this case, the tensor product of these operators is of type $(A \otimes B) : (V \otimes W) \mapsto (V' \otimes W')$.

In quantum mechanics, it is very common to use operators in the form of external products, for example, $A = |v\rangle\langle v|$ and $B = |w\rangle\langle w|$. The tensor product of A by B can be represented by the following equivalent ways:

$$\begin{aligned} A \otimes B &= (|v\rangle\langle v|) \otimes (|w\rangle\langle w|) \\ &= |v\rangle\langle v| \otimes |w\rangle\langle w| \\ &= |v, w\rangle\langle v, w|. \end{aligned}$$

If A_1, A_2 are operators in V and B_1, B_2 are operators in W , then the composition or the matrix product of the matrix representations obey the property

$$(A_1 \otimes B_1) \cdot (A_2 \otimes B_2) = (A_1 \cdot A_2) \otimes (B_1 \cdot B_2). \quad (26)$$

The inner product of $|v_1\rangle \otimes |w_1\rangle$ by $|v_2\rangle \otimes |w_2\rangle$ is defined as

$$(|v_1\rangle \otimes |w_1\rangle, |v_2\rangle \otimes |w_2\rangle) = \langle v_1 | v_2 \rangle \langle w_1 | w_2 \rangle. \quad (27)$$

The inner product of vectors written as a linear combination of basis vectors are calculated by applying the linear property in the second argument and the *conjugate-linear* property in the first argument of the inner product. For example,

$$\left(\left(\sum_{i=1}^n a_i |v_i\rangle \right) \otimes |w_1\rangle, |v\rangle \otimes |w_2\rangle \right) = \left(\sum_{i=1}^n a_i^* \langle v_i | v \rangle \right) \langle w_1 | w_2 \rangle.$$

The inner product definition implies that

$$\| |v\rangle \otimes |w\rangle \| = \| |v\rangle \| \cdot \| |w\rangle \|.$$

In particular, the norm of the tensor product of unit-norm vectors is a unit-norm vector.

When we use matrix representations for operators, the tensor product can be calculated explicitly via the *Kronecker product*. Let A be a $m \times n$ matrix and B a $p \times q$ matrix. Then,

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1n}B \\ & \ddots & \\ a_{m1}B & \cdots & a_{mn}B \end{bmatrix}. \quad (28)$$

The dimension of the resulting matrix is $mp \times nq$. The Kronecker product can be used for matrices of any dimension, particularly for two vectors,

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \otimes \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} a_1 \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \\ a_2 \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} a_1 b_1 \\ a_1 b_2 \\ a_2 b_1 \\ a_2 b_2 \end{bmatrix}.$$

The tensor product is an associative and distributive operation, but noncommutative, that is, $|v\rangle \otimes |w\rangle \neq |w\rangle \otimes |v\rangle$ if $v \neq w$. Most operations on a tensor product are performed term by term, such as

$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger.$$

If both operators A and B are special operators of the same type, as the ones defined in Sect. 3.4, then the tensor product $A \otimes B$ is also a special operator of the same type. For example, the tensor product of Hermitian operators is a Hermitian operator.

The trace of a Kronecker product of matrices is

$$\text{tr}(A \otimes B) = \text{tr } A \text{ tr } B,$$

while the determinant is

$$\det(A \otimes B) = (\det A)^m (\det B)^n,$$

where n is the dimension of A and m of B .

The direct sum of a vector space V with itself n times is a particular case of the tensor product. In fact, a matrix $A \oplus \cdots \oplus A$ in $V \oplus \cdots \oplus V$ is equal to $I \otimes A$ for any A in V , where I is the $n \times n$ identity matrix. This shows that, somehow, the tensor product is defined from

the direct sum of vector spaces, analogous to the product of numbers which is defined from the sum of numbers. However, the tensor product is richer than the simple repetition of the direct sum of vector spaces. Anyway, we can continue generalizing definitions: It is natural to define tensor potentiation, in fact, $V^{\otimes n}$ means $V \otimes \cdots \otimes V$ with n terms.

If the *diagonal state* of the vector space V is $|D\rangle_V$ and of space W is $|D\rangle_W$, then the diagonal state of space $V \otimes W$ is $|D\rangle_V \otimes |D\rangle_W$. Therefore, the diagonal state of space $V^{\otimes n}$ is $|D\rangle^{\otimes n}$.

Proof of the mixed-product property (26). Both sides are linear operators on $V \otimes W$, so it is enough to compare them on the basis vectors $|v\rangle \otimes |w\rangle$. Applying (25) twice,

$$(A_1 \otimes B_1)(A_2 \otimes B_2)(|v\rangle \otimes |w\rangle) = (A_1 \otimes B_1)(A_2 |v\rangle \otimes B_2 |w\rangle) = A_1 A_2 |v\rangle \otimes B_1 B_2 |w\rangle,$$

which is $((A_1 A_2) \otimes (B_1 B_2))(|v\rangle \otimes |w\rangle)$. $\square$

Corollary 4.3. $U \otimes U'$ is unitary whenever U and U' are, since by (26) and the dagger rule $(U \otimes U')^\dagger (U \otimes U') = (U^\dagger U) \otimes (U'^\dagger U') = I \otimes I = I$. Hence applying gates to disjoint sets of qubits in parallel is again a legitimate quantum operation.

Example 4.4 (Kronecker product of two gates).

$$X \otimes I = \begin{bmatrix} 0 \cdot I & 1 \cdot I \\ 1 \cdot I & 0 \cdot I \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad H^{\otimes 2} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}.$$

The operator $X \otimes I$ flips the first qubit and leaves the second alone; applied to $|0, 0\rangle$ it gives $|1, 0\rangle$, as reading its first column confirms. And $H^{\otimes 2} |0, 0\rangle = \frac{1}{2}(|0, 0\rangle + |0, 1\rangle + |1, 0\rangle + |1, 1\rangle) = |D\rangle$: two Hadamards prepare the uniform superposition over four inputs, the case $n = 2$ of Exercise A.15.

Pro Memoria: product states, entangled states, and exponential growth

The tensor product is where quantum computation departs from classical intuition. A state of two qubits that can be written as $|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$ is called a *product state*: each qubit has a state of its own. A state that cannot is called *entangled*. The smallest example is

$$|\Phi^+\rangle = \frac{|0, 0\rangle + |1, 1\rangle}{\sqrt{2}},$$

which is not a product state: matching $(\alpha_1 |0\rangle + \beta_1 |1\rangle) \otimes (\alpha_2 |0\rangle + \beta_2 |1\rangle)$ against it would require $\alpha_1 \alpha_2 = \beta_1 \beta_2 = 1/\sqrt{2}$ and $\alpha_1 \beta_2 = \beta_1 \alpha_2 = 0$, which is impossible. Note the counting: a product state of n qubits is specified by n pairs of amplitudes, i.e. $O(n)$ numbers, whereas a general state of n qubits needs 2^n amplitudes, as the next section makes explicit. Entangled states are precisely those that use that extra room, and the exponential gap between 2^n and $2n$ is the resource quantum algorithms try to exploit.

4.3 Registers

A *register* is a set of qubits treated as a composite system. In many quantum algorithms, the qubits are divided into two registers: one for the main calculation from where the result comes out and the other for the draft (calculations that will be erased). Suppose we have a register with two qubits. The computational basis is

$$|0, 0\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad |0, 1\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad |1, 0\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad |1, 1\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

A generic state of this register is

$$|\psi\rangle = \sum_{i=0}^1 \sum_{j=0}^1 a_{ij} |i, j\rangle$$

where coefficients a_{ij} are complex numbers that satisfy the constraint

$$|a_{00}|^2 + |a_{01}|^2 + |a_{10}|^2 + |a_{11}|^2 = 1.$$

To help generalizing to n qubits, it is usual to compress the notation by converting binary-base representation to decimal-base. The computational basis for two-qubit register in decimal-base representation is $\{|0\rangle, |1\rangle, |2\rangle, |3\rangle\}$. In the binary-base representation, we can determine the number of qubits by counting the number of digits inside the *ket*, for example, $|011\rangle$ refers to three qubits. In the decimal-base representation, we cannot determine what is the number of qubits. This information should come implicit. In this case, we can go back, write the numbers in the binary-base representation and explicitly retrieve the notation. In the compact notation, a generic state of a n -qubit register is

$$|\psi\rangle = \sum_{i=0}^{2^n-1} a_i |i\rangle, \quad (29)$$

where coefficients a_i are complex numbers that satisfy the constraint

$$\sum_{i=0}^{2^n-1} |a_i|^2 = 1.$$

The *diagonal state* of a n -qubit register is the tensor product of the diagonal state of each qubit, that is, $|D\rangle = |+\rangle^{\otimes n}$.

Remark 4.5 (the two notations side by side). For $n = 3$ the same basis vector may be written $|1, 0, 1\rangle$, $|101\rangle$ or $|5\rangle$, and the same state may be written $|\psi\rangle = \sum_{x \in \{0,1\}^3} a_x |x\rangle$ or $|\psi\rangle = \sum_{i=0}^7 a_i |i\rangle$. The binary form is the honest one, it displays the register structure and which qubit a gate acts on, while the decimal form keeps formulas short. Only the binary form makes the size of the register visible, so whenever a decimal label appears, the number of qubits must be stated somewhere.

Example 4.6. Expanding $|+\rangle^{\otimes n}$ with bilinearity gives

$$|D\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right)^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{i=0}^{2^n-1} |i\rangle,$$

in agreement with (5) for $n \rightarrow 2^n$: n Hadamard gates, acting in parallel on $|0\rangle^{\otimes n}$, produce a uniform superposition over all 2^n inputs, and by the corollary above the whole layer $H^{\otimes n}$ is unitary. This one line is the first step of nearly every quantum algorithm.

Where this leads

With (29) in hand, the standard structure of a quantum algorithm can now be stated in the language of these two lectures:

1. Prepare $|D\rangle = H^{\otimes n} |0\rangle^{\otimes n}$, a uniform superposition over 2^n inputs (Sections 1.4, 4.2).
2. Evaluate the function of interest with the reversible operator $U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$

on two registers (Exercise A.16); by Exercise A.16(1) this is unitary for every f .

3. Interfere, by applying further unitaries so that amplitudes of wrong answers cancel and the right one grows, arithmetic on the complex amplitudes a_i , made tractable by diagonalising the operator involved (Sections 3.1, 4.1).
4. Measure, obtaining outcome i with probability $|a_i|^2 = \langle \psi | P_i | \psi \rangle$ (Block 2 and Section 3.4).

Every step is one of the operations assembled in these two lectures. What distinguishes a good algorithm from a useless one is entirely step 3.

Exercises for this block: Exercise A.14 (matrix square roots), Exercise A.15 (the n -qubit Hadamard), Exercise A.16 (implementing a function reversibly); supplementary Exercise S.10 (Kronecker products and mixed products), Exercise S.11 (traces and expectation values), Exercise S.12 (product versus entangled states).